

3. The table shows how the solubility of potassium nitrate in water varies with temperature.

Temperature (°C)	Solubility (g per 100g of water)
0	13
10	21
20	32
30	46
40	64
50	86

(a) Tick (✓) the box that shows approximately how many times more soluble potassium nitrate is at 40 °C than it is at 10 °C.

[1]

2 times

☐

3 times

☐

4 times

☐

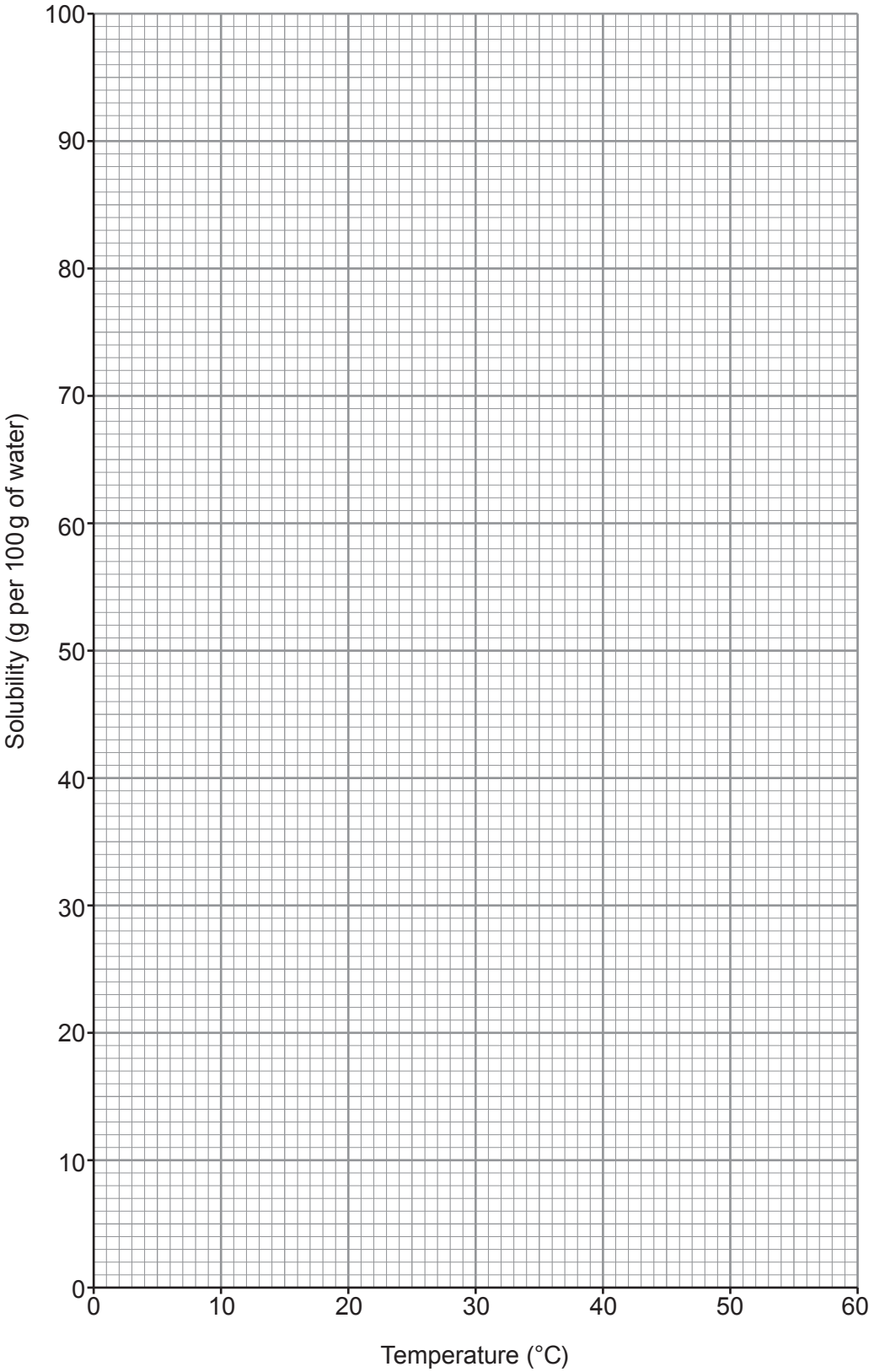
5 times

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Examiner
only

- (b) (i) Plot the solubility of potassium nitrate against temperature on the grid.
Draw a suitable line.

[3]



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- (ii) I. State the solubility of potassium nitrate at 45 °C. [1]

..... g per 100 g of water

- II. Calculate the mass of solid potassium nitrate that forms when a saturated solution in 100 g of water at 45 °C cools to 25 °C. [1]

Mass = g

- (c) Potassium nitrate decomposes on heating to produce potassium nitrite, KNO_2 , and oxygen.

- (i) Balance the equation for the reaction. [1]



- (ii) On heating potassium nitrate, a student expected to make 4.35 g of potassium nitrite. However, in an experiment she only made 3.13 g.

Calculate the percentage yield of potassium nitrite in her experiment. Give your answer to the nearest **whole** number. [2]

Percentage yield = %

